TEXORA · LECTURE SHEET

Production Systems in Apparels

The ways sewing work is organised — make-through, progressive bundle, unit production and modular team systems — compared on work-in-process, throughput time, quality feedback and the order profiles each system serves.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 2 · LEVEL 4 TERM 2

1. What a production system decides

A production system answers four questions for the sewing floor: **who** performs the operations (one operator per operation, or a team sharing them), **in what batch** work moves (bundles of 20–60 pieces, or single units), **how** it moves (trolley, conveyor, overhead hanger), and **where quality is caught** (at a distant end-of-line table, or inside the flow). These choices fix the factory's work-in-process (WIP), throughput time, defect feedback speed and changeover behaviour more than any machine purchase does. The systems below are points on a spectrum from batch to flow.

2. The main systems

SYSTEM	HOW IT WORKS	STRENGTHS / WEAKNESSES
Make-through (whole garment)	One skilled operator sews the complete garment start to finish.	Highest ownership and flexibility for samples, tailoring, tiny runs. Very low output per operator on volume styles; quality varies by operator; impossible to balance.
Progressive bundle system (PBS)	Operations are split along a line; bundles (tied groups of cut parts, typically 20–60 garments) travel station to station; operators specialise.	The industry default: simple, cheap, high specialisation efficiency. Weaknesses: huge WIP, throughput times of days, defects discovered bundles later, bundle handling and tracking overhead, line balancing by bundle juggling.

Unit production system (UPS)	Single garments travel on hangers along an overhead rail/conveyor with computer-controlled routing to stations; each operator receives one unit at a time.	Minimal WIP, throughput in hours, instant defect feedback, automatic production data, easy style change. Weaknesses: capital cost, maintenance, hanger-incompatible products (very large/small goods), dependence on the control system.
Modular / team system (quick response manufacturing)	Small teams (6–15 operators) own a garment section or whole garment, stand/move between machines, balance themselves, check quality in-team, paid often on team output.	Very fast throughput and changeover, built-in quality, multi-skilling and motivation. Needs trained, stable teams and strong team leaders; efficiency accounting is at team level; scaling to very long runs adds little.
Straight-line / conveyor variants	PBS or single-piece flow arranged on powered conveyors, sometimes with timed pacing.	Removes bundle carrying, paces the line. Still inherits the balance problems of its parent system.

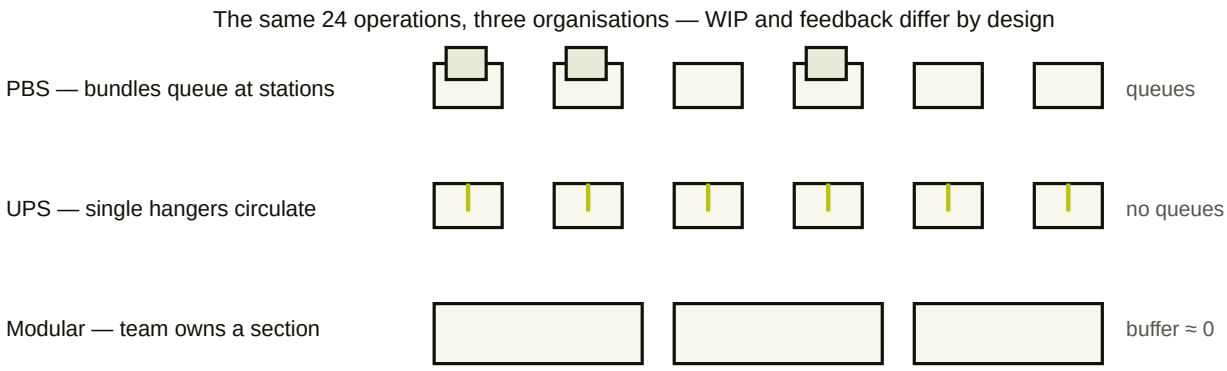


Figure 1 — System geometry: PBS stacks bundles before stations (small squares), UPS moves one hanger at a time (green drops), modular teams group stations so work is handed, not queued.

3. The governing arithmetic: Little's law

WORKED EXAMPLE — VERIFY EVERY NUMBER

Little's law: throughput time = WIP ÷ output rate. A PBS line holds WIP of 2,400 garments (bundles in queues, on machines, in transit) and ships 600 garments/day → throughput time = 2,400 ÷ 600 = **4 days** from cut to finish. The same line converted to single-piece flow holds WIP of 240 garments at the same 600/day → 240 ÷ 600 = 0.4 day = **3.2 working hours**. Nothing about operator speed changed: the 10× difference in response time, defect feedback and order lead time is pure system design. A defect created at operation 3 is found 4 days later in PBS (thousands of garments suspect) and within the hour in flow production (a handful). This is why quick-response buyers pay for flow systems.

4. Choosing the system

ORDER PROFILE	SYSTEM THAT FITS	WHY
Long runs, stable basics, price-driven	PBS (well balanced)	Specialisation efficiency outweighs WIP cost; throughput time tolerable.
Medium runs, frequent style change, branded basics	UPS	Changeover without re-balancing bundles; data for IE control (Topic 3).
Short runs, fashion, many styles per season	Modular teams	Team self-balancing absorbs style change; speed to market is the product.
Samples, couture, uniforms one-offs	Make-through	Batch of one; skill already paid for.

Real factories mix systems — PBS for the trouser plant, modular cells for fashion tops, UPS in the flagship line — and the planning system of Topic 4 must know each line's system, because WIP allowances and lead times in the plan are system properties.

Glossary

TERM	MEANING
Bundle	A tied batch of cut parts for a set number of garments, the transport unit of PBS.
WIP	Work in process: garments started but not finished, wherever they wait.
Throughput time	Elapsed time from a garment's start (cutting) to finish; = WIP ÷ output rate (Little's law).
UPS	Unit production system: computer-routed overhead hanger transport of single garments.

Modular system	Team-based production where a small group owns a garment/section and self-balances.

Key points and summary

- The production system fixes batch size, transport, WIP and feedback speed — the factory's response characteristics.
- PBS is cheap and specialised but carries days of WIP; UPS and modular systems buy hours-level throughput with capital or team skill.
- Little's law quantifies it: throughput time = WIP ÷ output rate.
- Choose by order profile: run length and style-change frequency decide.

Practice

Answer in your own words; brief model answers follow.

1. A line ships 450 garments/day and holds WIP of 1,800. What is its throughput time? WIP is halved by a flow conversion at the same output — what happens to throughput time and to the number of garments suspect when a defect is found at final check?
2. Why does PBS make line balancing harder than UPS, even with the same operations?
3. A buyer demands 5-day order-to-ship on fashion styles with weekly style changes. Which system do you propose and why?
4. Give two reasons a factory might keep make-through capacity even in a volume plant.
5. Why is defect feedback speed a financial variable, not only a quality one?

Answer key

1. Throughput time = 1,800 ÷ 450 = 4 days. Halving WIP to 900 → 2 days. Suspect garments on a defect finding scale with the work produced since the fault began — roughly the throughput window — so halving throughput time roughly halves the suspect population and the sorting cost.
2. In PBS, work arrives in bundles: an operator cannot start until a bundle arrives and cannot smooth imbalance smaller than a bundle; queues hide the true bottleneck. In UPS, single units arrive continuously, stations can be re-tasked in real time, and the control system shows true station loads minute by minute.
3. Modular teams (or UPS). Weekly changeovers destroy PBS economics — re-balancing and bundle learning curves repeat every style — while teams self-balance and throughput of hours, not days, makes a 5-day promise feasible.

4. Sample and development sewing (speed of one-off work, pattern feedback) and repair/complex bespoke items; also as a training ground where operators learn whole-garment construction before specialising.
5. Because late discovery converts a one-operation fault into lot-level sorting, rework, possible audit failure (AM-4102 Study 3) and shipment risk. The cost of a defect grows with the WIP between its creation and its detection — a cash figure, computable per system.
